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4. Poincaré in "The Nature of Mathematical Reasoning" cites the principle of mathematical induction, which he also calls proof by recurrence, as the element in mathematical reasoning that goes beyond mere logical deduction and raises mathematics to the level of a science. How would Hempel [in "The Nature of Mathematical Truth"], disagree? Would his reasons for disagreement be good ones?

In Henri Poincaré's "The Nature of Mathematical Reasoning"[1] he examines the possibility of classifying mathematics as a science due to its apparent use of inductive reasoning. In analyzing proofs of basic mathematical concepts such as natural numbers and operations, Poincaré explains the concept of *proof by recurrence*, in which a pattern of reasoning extends infinitely in order to establish a general rule. Proof by recurrence works by proving a base case and then showing that if a statement holds for one number, it must hold for the next. Eventually, the verification of the statement for each number need not continue, and it follows that the statement will continue to hold with certainty.

Poincaré likens the concept of proof by recurrence to scientific induction as it is used to translate experimental results of a few particular observations into generalized laws of science. He states: "Mathematics may, like the other sciences, proceed from the particular to the general" and deems proof by recurrence analogous to ordinary induction."[2] That is, if a certain finite number of iterations of a pattern can be verified, the pattern in general can be deemed to hold infinitely. He points out that this mirrors the way in which observing physical phenomena can lead us to conjecture generalized physical rules. For example, a physicist's repeated observations of instances where that f = m*a might suggest to her that such a relation is a law of physics that will hold in all cases, even though she has only observed it a finite number of times.

[1]Benacerraf, Paul, and Hilary Putnam, editors. *Philosophy of Mathematics: Selected Readings*. 2nd ed. Cambridge: Cambridge University Press, 1984, pp. 394-402.

[2]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 401.

In Carl Hempel's "The Nature of Mathematical Truth"[3] he presents arguments that are contradictory to the analogy between mathematics and science put forth by Poincaré. For the remainder of this paper, I will discuss two potential arguments against Poincaré's comparison: first due to the differing levels of certainty in mathematical and scientific conclusions, and second due to the way in which science reflects the physical world whereas mathematics does not. Then, I will evaluate how well they challenge Poincaré's analogy between inductive reasoning in mathematics and the sciences

Hempel might first point out that proof by recurrence and inductive reasoning in physical science result in very different degrees of confidence in their conclusions. He states: "While the statements of empirical science are constantly subject to revision in light of new evidence, the truth of an analytic statement can be established definitely, once and for all."[4] Thus, we consider proof by recurrence to be a rigorous basis for proof, while conclusions drawn from experimentation in physical sciences seem far less certain. Hempel explains that we can conceive of finding evidence that disproves a physical law, perhaps the planets reversing in their orbits. However, we cannot conceive of a type of evidence that would suggest that the conclusions of mathematical proofs are not steadfast, including those that rely on mathematical induction.[5] That is, it seems plausible that f = m*a could one day be demonstrated false as long as there is evidence to suggest otherwise. On the other hand, it does not seem remotely possible that 2 + 3 would someday be discovered as resulting in anything other than 5. Thus, Hempel might argue that mathematical results, including those generated by inductive reasoning, are intuitively irrefutable; this quality of theirs diverges from the flexible and consistently evolving laws of physics. Thus, the results of mathematical induction are inconsistent with the results of scientific induction in their degree of certainty.

[3]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 377-393.
[4]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 379.
[5]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 378.

Poincaré also asserts that the use of inductive reasoning in proof by recurrence extends beyond the logical deductive reasoning that mathematics uses, and introduces an aspect of drawing conclusions from observation as done in science. In response to this, Hempel might look to the sources of these observations for mathematics and sciences respectively. In science, all knowledge comes from observation of the physical world. For example, inductive reasoning could be based on an observation of a particle or celestial body. According to Hempel, mathematics does not reflect an external reality in the same way as hard sciences; pure mathematics is distinct from the outside world.[6] In Hempel's view, mathematical reasoning is self-evident, akin to a formal extension of logical reasoning, where the entire body of knowledge can be deduced from itself.[7] In this way, math is as immutable as the logical axioms and definitions that construct it, and every conclusion possibly drawn is implicit in the axioms already set at the outset. This is in contrast to physics, where new observations are necessary for some new discoveries. For example, if scientists observed two particles interacting in a way previously not seen, they would learn something new about physics that could not have possibly been ascertained with their body of knowledge prior to the discovery. In mathematics, there is nothing to be observed, new knowledge is computed using preexisting axioms in different combinations and orders. In physical sciences, the discovery of a new particle or behavior could lead to newness that is simply nonexistent in mathematics.

Hempel does acknowledge that, depending on how logic is defined, mathematics extends logic with the addition of defined concepts, such as numbers. One might argue that because numbers are so prevalent and deeply embedded in our descriptions of the physical world, mathematical reasoning could be considered empirical in some sense, similar to scientific reasoning. However, Hempel might combat this with his arguments from Section 5, where he emphasizes that mathematics is still a deductive system, and all mathematical propositions are logically derived. While numbers may appear to add an empirical dimension, they are merely

[6]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 378.
[7]Benacerraf and Putnam, *Philosophy of Mathematics*, pp. 377.

formal constructs whose properties are dictated by stipulated definitions rather than external observation. Thus, 2 + 3 = 5 is not an empirical fact but follows necessarily from the definitions of 2, 3, and 5, along with the stipulated rules of addition.

Now I will evaluate the effectiveness of Hempel's arguments against mathematics as a science. Poincaré's original statement is notably brief; he does not argue that the processes of proof by recurrence are equivalent to scientific induction but simply demonstrates the ways in which they are analogous. Therefore, Hempel's counterpoints must threaten the strength of this analogy, not just dissect the fine details of any divergences. To his first point on the differing levels of certainty of mathematical and scientific results, this difference does not necessarily undermine the analogy between mathematical induction and scientific induction. In both cases, a pattern established through particular and finite instances is extended to an infinite or universal claim. Poincaré could counter that the key similarity lies in the structure of reasoning rather than the certainty of the conclusions. While scientific laws remain subject to revision, and mathematical theorems proven by induction do not, this does not require that the reasoning behind them operates on fundamentally different principles, only that mathematics operates within a stricter system.

A similar argument could be made for the second Hempel argument: that scientific induction is based on observation of the physical world, whereas mathematical induction is purely formal and does not require empirical verification. Once again, this distinction does not negate the structural similarities between the two kinds of induction. However, the way in which mathematical conclusions are predetermined by the axioms and definitions established at the outset does present a challenge to Poincaré's analogy. In science, induction generates new knowledge about the world; observations of planetary motion, for example, inform physical laws that were not previously known. In contrast, mathematical induction does not reveal new truths in the same way but rather exposes what was already implicit within the body of knowledge set by the axioms and definitions of the system. This difference weakens

Poincaré's argument by suggesting that while mathematical induction may resemble scientific induction in form, it does not play the same epistemic role in expanding human knowledge; it only deduces what is already immutable. That is, the way in which proof by recurrence guarantees the continuation of the pattern is not just from our recognition of a continuing pattern, but from our confidence in mathematics being predetermined, and therefore predictable.

Ultimately, while Poincaré puts forth a compelling analogy between mathematical induction and scientific induction, the reasoning found in proof by recurrence fails to deem mathematics an inductive science. Hempel's second argument concerning the self-evident nature of mathematical reasoning is quite effective in refuting Poincaré's assertion that proof by recurrence is akin to the inductive reasoning used in sciences; given that the body of mathematical knowledge is predetermined according to its axioms and definitions, it is impossible for its reasoning to be anything other than deductive, even though the structure of reasoning used might appear similar to that of induction.